King Fahd University of Petroleum and Minerals

Department of Mathematics

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Math 101 Major Exam I Term 231

Original exam date: not printed. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

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1. $\lim_{x\to\infty}\left(x-\sqrt{x^2+x}\right)=$

(a) $-\infty$

(b) 0

(c) $\dfrac{1}{2}$

(d) $-\dfrac{1}{2}$

(e) -1

2. $\lim_{t\to 0}\dfrac{\sin(6t)}{2t}=$

(a) ∞

(b) 3

(c) 6

(d) $\dfrac{1}{2}$

(e) 1

3. The graph of $f(x)=\dfrac{x^2+2x-8}{x^3-4x}$ has (V. A.: vertical asymptote; H. A.: horizontal asymptote)

(a) 3 V.A and 1 H. A.

(b) 2 V.A and 1 H. A.

(c) no V.A and 1 H. A.

(d) 1 V.A and 2 H. A.

(e) no V.A and 2 H. A.

4. Find the derivative from the left of $f(x)=|x-1|$ at $x=1$.

(a) ∞

(b) 1

(c) 0

(d) -1

(e) $-\infty$

5. Which one of the following statements is TRUE about $f(x) = \dfrac{x}{x^2 - x}$?

(a) f has a nonremovable discontinuity at $x = 0$.

(b) f has a nonremovable discontinuity at $x = 2$.

(c) f has a removable discontinuity at $x = -1$.

(d) f has a removable discontinuity at $x = 0$.

(e) f has a removable discontinuity at $x = 1$.

6. $\lim_{x \to -\infty} \dfrac{5x^3 + 1}{10x^3 - 3x^2 + 7} =$

(a) 5

(b) $-\dfrac{1}{2}$

(c) $-\infty$

(d) 0

(e) $\dfrac{1}{2}$

7. If $f(x) = 6\sqrt{x} + 5\cos x + 11$, then

(a) $f'(x) = \dfrac{-3}{\sqrt{x}} - 5\sin x$

(b) $f'(x) = 2\sqrt{x} + 5\sin x$

(c) $f'(x) = \dfrac{3}{\sqrt{x}} - 5\sin x$

(d) $f'(x) = \dfrac{6}{\sqrt{x}} - 5\cos x$

(e) $f'(x) = \dfrac{3}{\sqrt{x}} + 5\sin x$

8. $\lim_{x \to -5^-} \dfrac{x + 3}{x^2 + 8x + 15} =$

(a) $-\infty$

(b) 0

(c) 1

(d) ∞

(e) 3

9. $\lim_{x\to -1^-}\left(\left\lfloor \frac{x}{3}\right\rfloor - \lfloor x\rfloor\right) =$ ($\lfloor\cdot\rfloor$ is the greatest integer function).

(a) -1

(b) 2

(c) 0

(d) 1

(e) $\frac{2}{3}$

10. The Intermediate Value Theorem guarantees that the function $f(x) = x^3 - x^2 + 2x - 3$ has a zero in the interval

(a) $[2, 3]$

(b) $[-1, 0]$

(c) $[-1, 1]$

(d) $[0, 1]$

(e) $[0, 2]$

11. If the line $y = -\frac{3}{4}x + 3$ is tangent to the graph of $f(x) = \frac{k}{x}$, then

(a) $k = -\frac{1}{3}$

(b) $k = 3$

(c) $k = -\frac{3}{4}$

(d) $k = 1$

(e) $k = -\frac{2}{3}$

12. $\lim_{x\to 2}\frac{x^2 + 2x - 8}{x^2 - x - 2} =$

(a) -3

(b) 0

(c) 4

(d) 2

(e) 1

13. If $f(x) = \begin{cases} 1 - x^2, & x \leq c \\ x, & x > c \end{cases}$ is continuous at $x = c$, then $(2c + 1)^2 =$

(a) 3

(b) 1

(c) 7

(d) 5

(e) 9

14. $\lim_{x \to 3} \dfrac{\sqrt{x+1} - 2}{x - 3} =$

(a) $\dfrac{1}{4}$

(b) $\dfrac{1}{2}$

(c) 0

(d) $\dfrac{2}{3}$

(e) $\dfrac{-1}{3}$

15. The function $f(x) = \dfrac{\sqrt{x+3}}{x}$ is continuous on

(a) $[-3, 0) \cup (0, \infty)$

(b) $[-3, \infty)$

(c) $(-\infty, 0) \cup (0, \infty)$

(d) $(-\infty, \infty)$

(e) $(-\infty, -3]$

16. $\lim_{x \to 0} \dfrac{2 - 3\tan(3x) - 2\cos x}{2x} =$

(a) -3

(b) $\dfrac{3}{2}$

(c) $-\dfrac{9}{2}$

(d) -1

(e) 3

17. If $y = Ax + B$ is an equation of the tangent line to the graph of $f(x) = \dfrac{2}{\sqrt[4]{x^3}}$ at the point $(1, 2)$, then $B - A =$

(a) -2

(b) -3

(c) $-\dfrac{5}{2}$

(d) 4

(e) 5

18. If $f(x) = \begin{cases} x^2, & x \leq 2 \\ 8 - 2x, & 2 < x < 4 \\ 4, & x \geq 4 \end{cases}$ then $\lim_{x \to 2} f(x)$

(a) is equal to 1

(b) is equal to 4

(c) does not exist

(d) is equal to 6

(e) is equal to 2

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	Conjugate: $\frac{-x}{x+\sqrt{x^2+x}} = \frac{-1}{1+\sqrt{1+1/x}} \to -\frac{1}{2}$.
2	**(b)**	$\frac{6}{2} \cdot \frac{\sin 6t}{6t} \to 3 \cdot 1 = 3$.
3	**(b)**	Simplified $\frac{x+4}{x(x+2)}$: V.A. $x = 0, -2$; H.A. $y = 0$ only.
4	**(d)**	For $x < 1$: $\frac{-(x-1)}{x-1} = -1$, so $f'_-(1) = -1$.
5	**(d)**	$f = \frac{1}{x-1}$ for $x \neq 0$: limit -1 at 0 (removable); $x = 1$ is infinite.
6	**(e)**	Divide by x^3: limit is $\frac{5}{10} = \frac{1}{2}$.
7	**(c)**	Power rule and $(\cos x)' = -\sin x$: $f' = \frac{3}{\sqrt{x}} - 5\sin x$.
8	**(a)**	Simplifies to $\frac{1}{x+5}$, and $x + 5 \to 0^-$: limit $-\infty$.
9	**(d)**	$\lfloor x/3 \rfloor \to -1$, $\lfloor x \rfloor \to -2$, so $-1 + 2 = 1$.
10	**(e)**	Only $f(0) = -3 < 0 < 5 = f(2)$ changes sign.
11	**(b)**	Slope and height conditions give $a = 2$, so $k = \frac{3}{4}a^2 = 3$.
12	**(d)**	Factor and cancel $(x - 2)$: $\frac{x+4}{x+1} \to \frac{6}{3} = 2$.
13	**(d)**	$1 - c^2 = c \Rightarrow c^2 + c = 1$, so $(2c + 1)^2 = 4(1) + 1 = 5$.
14	**(a)**	Rationalize: $\frac{x-3}{(x-3)(\sqrt{x+1}+2)} \to \frac{1}{4}$.
15	**(a)**	Need $x \geq -3$ and $x \neq 0$: $[-3, 0) \cup (0, \infty)$.
16	**(c)**	$\frac{1-\cos x}{x} \to 0$ and $\frac{3\tan 3x}{2x} \to \frac{9}{2}$, so $-\frac{9}{2}$.
17	**(e)**	$A = f'(1) = -\frac{3}{2}$, $B = 2 + \frac{3}{2} = \frac{7}{2}$, so $B - A = 5$.
18	**(b)**	Left limit $= 4$, right limit $= 8 - 4 = 4$, so the limit is 4.